ties cannot be treated lightly. Some decision theorists have taken the paradox quite seriously and have proposed modifications of utility theory to avoid it.

PROBLEMS

1. Ellsberg presented another paradox. If we offer people a bet on a fair coin, most are indifferent between betting on heads and betting on tails. They are similarly indifferent when offered a bet on biased coins when they are not told the bias. However, when offered the choice

   bet heads on the fair coin vs. bet tails on the biased one

   they prefer to bet on the fair coin. Suppose that whichever coin is involved in a winning bet pays $1, whereas a losing one pays nothing. Show that the choices most people make contradict the combined theories of utility and subjective probability.
2. Of course, if you are a fan of utility theory and have reservations about subjective probabilities, the Ellsberg paradox confirms your suspicions. Explain why.
3. In both the Allais and Ellsberg paradoxes we have appealed to what "most people" think or to what "many apparently rational and thoughtful people" think. What bearing, if any, does this have on decision theory, construed as a theory of ideally rational beings?
4. Utility theory originated in attempts to deal with a version of the St. Petersburg game played with monetary prizes. In 1730, Daniel Bernoulli proposed that monetary values be replaced with their logarithms. If we use logarithms to the base 10, then the expected value of this version of the St. Petersburg game is

   $$(1/2)\log_{10}(2) + (1/4)\log_{10}(4) + (1/8)\log_{10}(8) + \ldots,$$

   which converges to a finite sum. Notice, however, that $\log_{10}(10) = 1$, $\log_{10}(100) = 2$, $\log_{10}(1{,}000) = 3$, etc., so that we can reinstate the St. Petersburg paradox by increasing the payoffs. Give a detailed explanation of how this can be done.

4-5. The Predictor Paradox

Here is a paradox, introduced by a physicist named Newcomb, which has generated more controversy than all the others combined.

Suppose a being with miraculous powers of foresight, called *the Predictor*, offers you the following choice. There are two shoe boxes on the table between the two of you. One is red, the other blue. The tops are off the boxes and you can see that the red box has $1,000 in it and that the blue box is empty. The Predictor tells you that you are to leave the room and return in five minutes. When you return you will be offered the choice of having the contents of the blue box or that of both boxes. The Predictor also tells you that while you are out of the room he will place $1,000,000 in the blue box if he predicts that you will take just the blue box, and he will leave it empty if he predicts that you will take both boxes. You leave the room and return after five minutes. The top is

still off the red box and the $1,000 is still in it. The blue box is now closed. You have heard about the Predictor before. You know that almost everyone who has taken both boxes has received just $1,000 while almost everyone who has taken just the blue box has received $1,000,000. What is your choice?

Suppose you use decision theory to help with your choice and your utility for money is equal to its EMV. Then you will have decision table 4-7. The

4-7

	Blue Box Empty	Not Empty
Take Blue Box	0 .1	1M .9
Take Both Boxes	$1,000 .9	1M + $1,000 .1

probabilities are determined as follows: You reckon that if you take just the blue box there is a 90% chance that the Predictor will have predicted that you would and will have put $1,000,000 in that box. You also think that there is only a 10% chance that he will have predicted incorrectly and left the box empty. Similarly you reckon that if you take both boxes, there is a 90% chance that he will have predicted that and left the blue box empty. It is easy to see that, given this problem specification, you will maximize expected utility by taking just the blue box. Recalling all your rich friends who took just the blue box, you are about to reach for it.

But doubt strikes. You notice that the dominant choice is to take both boxes. What then should you do? Decision theory's two most venerable rules offer you conflicting prescriptions!

Then you remember that the dominance principle is supposed to yield to the principle of maximizing expected utility when the probabilities of one or more states are dependent on the choices made. This made good sense in the nuclear disarmament example because disarming made an attack by the other side highly likely. But does it make good sense in this case? You argue to yourself: The $1,000,000 is already in the blue box or it is not. My choice cannot change things now. If there is $1,000,000 in the blue box, then taking both will yield $1,001,000 which is better than taking just the blue box. If the $1,000,000 is not there, I can get $1,000 by taking both boxes, and that's better than taking just the blue box and getting nothing. So in either case I am better off taking both boxes.

Yet next you remember that your friends who reasoned in this way are still poor whereas those who took the blue box are rich. But that does not seem to make sense either. Can the Predictor exercise mind control and influence your choice, thereby making his prediction come true? But how? He has not even told you what his prediction is. Or can your choice somehow influence his prediction? But again how? You choose *after* he predicts.

The dilemma is this. If we use the dominance principle to decide, we must

ignore all the empirical data pointing to the folly of choosing both boxes; but if we follow the data and maximize expected utility, we are at a loss to explain why the data are relevant. So what is your choice?

If you do not know what to do, take comfort: You are not alone. Many decision theorists think you should take both boxes but many others think you should take just the blue one. Moreover, this paradox has shaken decision theory to its foundations.

Before we consider some of the growing number of responses to the Predictor paradox, let us bring it down to Earth. (Not that this fanciful example is not useful, for, even if it might not arise in practice, it points to a conceptual weakness in decision theory.) Here are two real-life problems closely related to the Predictor paradox.

The first is derived from the religious views of the sixteenth-century Protestant theologian John Calvin. According to one interpretation of Calvin's views, God has already determined who will go to Heaven and nothing we do during our lifetimes—whether we be saints or sinners—will change this. On the other hand, although no mortal can know whom God has chosen, Calvinists believe that there is a high correlation between being chosen and being devout. Now suppose you are a dyed-in-the-wool Calvinist and are sorely tempted to commit some sin. You consider two arguments. According to the first you should pick the dominant act. Go ahead and sin, since that is not going to affect your going to Heaven anyway, and an earthly life with pleasure is better than one of Calvinist abstinence. But the other argument tells you that you should restrain yourself. As a devout Calvinist, you are certain that your sinning would be a clear sign that you are not among the elect.

Now for a contemporary example. Smoking cigarettes is highly correlated with heart disease. Smokers are much more likely to have heart troubles than nonsmokers and heavy smokers are at greater risk than light ones. Heart disease also seems to run in families and to be associated with ambitious, serious, and competitive individuals, known as type A personalities. Today most people researching the causes of heart disease regard smoking, genetic predisposition, and the type A personality as independent contributing causes of heart disease. However, suppose the causal story went as follows instead: Type A personality is inherited (just as breeding has made race-horses high strung and draft horses calm), and people with type A personalities have a greater need to smoke than do other people. However, smoking does not cause heart disease, rather the cause is the same genetic defect that produces the type A personality. Thus the cause of smoking and heart disease is at bottom one and the same—a genetic defect. Let us suppose that the same story held for the other diseases—for example, cancer, emphysema—in which smoking has been implicated. If you believed this story and were trying to decide whether to smoke, you might be in a dilemma similar to the Predictor paradox. For the high correlation between smoking and various diseases bestows a higher expected utility on not smoking. But you would either have a disease-causing genetic defect or not. If you had it, smoking would not increase your risk of disease, and it would (or so I will suppose) lead